

Notes for Replit: feelzlike fixes (checked 24 Sep 2026, 3:30pm AEST)

The republish changed the front-end build (index-C9prnJ5K.js became index-CBbgAYNV.js). But none of the fixes below are live yet. Please make each change, deploy BOTH the front end and the API server, and confirm with the checks listed.

1. URGENT: Thredbo and Jindabyne still crash (HTTP 500)

- GET /api/weather/thredbo and GET /api/weather/jindabyne still return:
WEATHER_FETCH_ERROR · Expected integer, received float · path hourly[n].cloudCover
- Cause: lib/api-spec/openapi.yaml declares cloudCover as type: integer in two schemas (around lines 2870 and 2985). The weather data now sends decimals.
- Fix:
 - Change both to type: number, then regenerate lib/api-zod and lib/api-client-react.
 - Also Math.round() cloudCover in artifacts/api-server/src/routes/weather.ts before validation, as a belt-and-braces measure.
 - Look for any other type: integer weather fields (humidity, visibility, wind direction) that could fail the same way.
- Make sure the API server deployment is republished, not just the front end.
- Check: both URLs return 200, and <https://feelzlike.com/snowy-mountains/mountain/thredbo/> shows conditions.

2. Snow forecast elevation is too low for 247 of 374 US/Canada mountains

- midMountainElevation(summitM) in artifacts/feelzlike/src/lib/elevation.ts treats elevationM as the summit and subtracts up to 300 m. But elevationM is often the base:
 - 30 authored US resorts store the base as elevationM. The server's weather.ts has the true summit. Examples: Jackson Hole 1924 vs 3185, Heavenly 1907 vs 3068, Breckenridge 2926 vs 3914, Alta, Mammoth, Palisades, Park City, Stowe, Killington.
 - 57 western-US catalogue records set forecastElevationM equal to baseElevationM, even though topElevationM exists.
 - 142 East/Midwest US catalogue records have no base or top, so the value is the map-pin ground height. Subtracting from it puts the forecast below the base lodge.
 - 18 (mostly Ontario and Prairie hills) land 10–40 m below a known base.
- Fix:
 - Add explicit baseElevationM and summitElevationM fields to MountainLink and retire the ambiguous elevationM.
 - Replace the snow elevation logic with:

```
export function snowForecastElevation(baseM?: number, summitM?: number, pinM?: number) {
  if (baseM && summitM && summitM > baseM) return Math.round(baseM + (summitM - baseM) * 0.5);
  if (summitM) return Math.max(baseM ?? 0, midMountainElevation(summitM));
  return baseM ?? pinM; // never subtract from a base or pin height
}
```

- Use it everywhere `midMountainElevation / Lb()` feeds `snowElevationM`: `MountainDetail.tsx`, `snowy-mountains LocationDetail.tsx` and `yamanouchi resort.tsx`.
- For the 30 authored resorts, copy the `summit` from `weather.ts` into each region file and keep the current value as the base.
- Add a test that fails if any snow elevation is below the base, or if client and server elevations differ by more than 150 m.
- Check: the Heavenly page should request about `snowElevationM=2488` (it's 1621 now), and Jackson Hole about 2555 (1635 now).
- The full list with suggested values is in `us_canada_snow_elevation_audit.csv`.

3. Internal research notes showing to the public

- The Jackson Hole blurb still shows "⚠ no confirmed dedicated webcam URL found in research, treat as unverified" and "2025-26" wording. It's in both `regions/jackson-hole.ts` and `api-server/src/routes/weather.ts`.
- The Snow King blurb shows a note about conflicting closing-date sources.
- Search all region files, catalogue data and `weather.ts` for "⚠", "unverified", "in research", "treat as", "per resort", "2025-26" and "tracker". Rewrite each one as customer-facing copy.

4. Mobile: two prompts overlap

- On a phone-sized screen, the "Add to home screen" and "Never miss a powder day" prompts appear at the same time, stacked with overlapping text.
- Show one at a time. Suggested order: the cookie banner first, then install, and powder alerts later in the session or on a second visit.

5. Units and labels

- Default units by region: °C/km/h for AU, NZ, JP and AT; °F/mph for the US. Remember the user's choice.
- Don't mix systems. With °F selected, rain still shows in mm and snow in cm.
- The 5-day forecast shows "Moderate Snow Fall" next to 0.1 cm. Base the label on the amount, not the weather code alone.
- The snowmaking card said "best window · thu 5am" at 9:50am on Thursday. That's likely a timezone/day-label bug and should probably read Fri.
- The blue "perisher.com.au" link on the blue header background has low contrast.

6. Reliability

- Firing a few hundred weather requests at once briefly returned 500 for 137 mountains, though every one worked when retried slowly. This was my audit traffic, not real users. Still, on a big powder morning the API should serve the last cached forecast instead of a 500. Consider caching per location, retry/backoff on upstream errors, and a `stale: true` flag like `/api/regions` already uses.

7. GitHub is out of date

- The GitHub repo `DarkHorse263/Yamanouchi-Weather-App` was last updated 7 Sep 2026. Please push the current Replit code to GitHub after these fixes, so future audits check the real code.